Ensemble_Methods

1. Bagging (Parallel)
    1. Considers homogeneous weak learners, learns them independently from each other in parallel and combines them following some kind of deterministic averaging process. Decrease variance.
2. Boosting (Sequential)
    1. Considers homogeneous weak learners, learns them sequentially in a very adaptive way (a base model depends on the previous ones) and combines them following a deterministic strategy. Decrease bias.
    2. XGBoost as a boosting method is a category of ensemble methods. Rather than training all of the models in isolation of one another, boosting trains models in succession, with each new model being trained to correct the errors made by the previous ones. Models are added sequentially until no further improvements can be made, that's why it is called additive model.
    3. XGBoost: One of the most powerful weapons in Kaggle. Many deep learner use it reached top in many competitions.